

INSTRUCTION CODES

Name	Assembly Command	Octal codes (if valid)		Register Field	Indicators Changed (LT & EQ)	Overflow can be set to 1
		Store Mode	Direct Mode			
Add	ADD	06	46	YES	YES	YES
And	AND	02	42	YES	YES	NO
Arithmetic Shift	ASH	24	64	YES	YES	NOTE 1
Call	CALL	10	-	NO	NO	NO
Change Process	CHP	11	51	NO	NOTE 2	
Compare	CMP	05	45	YES	YES	YES
Correlate	COR	15	55	YES	YES	NO
Cyclic Shift	CSH	26	66	YES	YES	NO
Divide	DIV	16	56	YES	YES	YES
Exclusive Or	EOR	00	40	YES	YES	NO
Unconditional Jump	JMP	36	76	7	NO	NO
Jump if Equal	JEQ	36	76	0	NO	NO
Jump if Greater than	JGT	36	76	2	NO	NO
Jump if Greater or Equal	JGE	36	76	4	NO	NO
Jump if Less than	JLT	36	76	3	NO	NO
Jump if Less or Equal	JLE	36	76	5	NO	NO
Jump if not Equal	JNE	36	76	1	NO	NO
Jump if Overflow	JOV	36	76	6	NO	NOTE 3
Load Capability	LC	30	70	YES	NO	NO
Load Data	LD	03	43	YES	YES	NO
Load Data Masked	LDM	23	63	YES	YES	NO
Load Negative	LDN	04	44	YES	YES	YES
Load Pointer	LDP	31	71	YES	YES	NO
Logical Shift	LSH	22	62	YES	YES	NO
Move	MOVE	-	74	NO	YES	YES
Multiply	MPY	17	57	YES	YES	YES
Or	OR	01	41	YES	YES	NO
Return	RET	-	75	NO	NO	NO
Store Capability	SC	34	-	YES	NO	NO
Store Data	SD	13	-	YES	YES	NO
Store Data Masked	SDM	33	-	YES	YES	NO
Subtract	SUB	07	47	YES	YES	YES
Swap	SWP	12	-	YES	YES	NO
Swap Masked	SWPM	32	-	YES	YES	NO

NOTE 1 Indicators Preserved

NOTE 2 YES if left shift otherwise NO

NOTE 3 Cleared

INSTRUCTION CODES (Cont'd)

SHIFT: +VE VALUES SHIFT LEFT
-VE VALUES SHIFT RIGHT

CORRELATE: SCANS FROM LEFT TO RIGHT

RESULT = 0 IF BIT 23 SET
= 1 IF BIT 22 SET
ETC.
= 32 IF NO BITS SET

MULTIPLY: DOUBLE LENGTH RESULT
LS HALF IN D REGISTER
MS HALF IN D + 1 REGISTER

DIVIDE: DOUBLE LENGTH DIVIDEND
LS HALF IN D REGISTER (23 BITS)
MS HALF IN D + 1 REGISTER (24 BITS)
RESULT IN D REGISTER
REMAINDER IN D + 1 REGISTER
INDICATORS SET ON RESULT

MOVE: E.G. MOVE D2 0 D5
SOURCE C2
SOURCE OFFSET D2
DESTINATION C3
DESTINATION OFFSET D3
SUMCHECK D5
COUNT D0
OVERFLOW MUST BE RESET BEFORE MOVE
ON EXIT, OVERFLOW SET BY 'STATUS' WIRE
INDICATORS SET ON COUNT
IF COUNT NON-ZERO, DEVICE IS 'BUSY'

23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
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STORE MODE INSTRUCTION

FUNCTION	REG	MOD	CAP	ADDRESS
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DIRECT MODE INSTRUCTION

FUNCTION	REG	MOD	SIGNED LITERAL
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COS

1	1	E	W	R	E	W	R	0
C	C	C	C	D	D	D	D	

CAPABILITY POINTER

POS

0	1	1	E	W	R	E	W	R
C	C	C	C	D	D	D	D	

ACCESS CODES